

Current Environmental Issues of Importance

Exponential growth of human population, rising standards of living, rapid industrialization, urbanization and addition of a large group of materials in the environment leading to problems like global climate change and acid rain have become important environmental issues in the current times.

7.1 POPULATION GROWTH

In 1800, the earth was home to about 1 billion people. The dramatic way in which global human population grew thereafter is shown in Fig. 7.1. It took about thirty nine thousand years of human history to reach 1 billion, 130 years to reach the second billion, 45 years to reach 4 billion and the next doubling is likely within a span of a few decades. We have already crossed 6 billion and may reach 11 billion by 2045 as per the World Bank estimates.

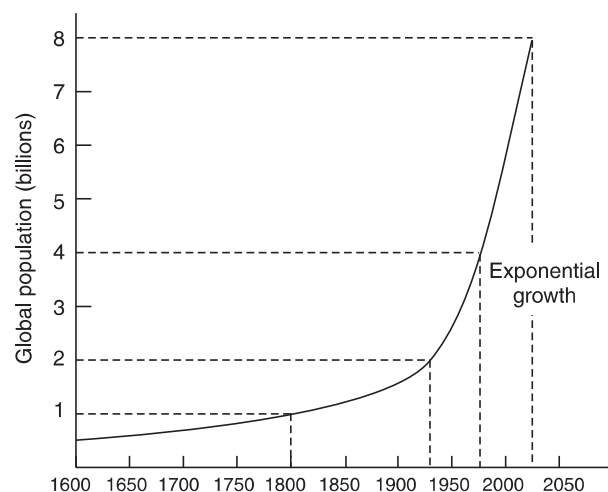


Fig. 7.1. Global population growth trends in the last four centuries.

Let us look at the reasons of this trend of human population growth. In the beginning of human civilization, during the Stone Age, population was quite stable. Environmental conditions were hostile

and humans had not yet developed adequate artificial means for adaptations to these stresses. Droughts and outbreak of diseases used to be quite common leading to mass deaths. The 14th century A.D. experienced large scale mortality due to bubonic plague when about 50% of people in Asia and Europe died due to the disease.

With scientific and technological advancement, life expectancy of humans improved. People started living in definite settlements leading a more stable life with better sanitation, food and medical facilities. Victory over famine-related deaths and infant mortality became instrumental for a rapid increase in population size. In agriculture based societies children were considered as economic assets who would help the parents in the fields and that is why in the developing countries, population growth climbed to unthought-of heights, at the rate of 3–4% per year, accounting for about 90–95% of total population growth of the world in the last 50 years.

(A) Population Characteristics and Variations Among Nations

Exponential growth: When a quantity increases by a constant amount per unit time *e.g.* 1, 3, 5, 7 etc. it is called linear growth. But, when it increases by a fixed percentage it is known as exponential growth *e.g.* 10, 10², 10³, 10⁴, or 2, 4, 8, 16, 32 etc. Population growth takes place exponentially and that explains the dramatic increase in global population in the past 150 years.

Doubling time: The time needed for a population to double its size at a constant annual rate is known as doubling time. It is calculated as follows:

$$\begin{aligned} T_d &= 70/r \\ \text{where } T_d &= \text{Doubling time in years} \\ r &= \text{annual growth rate} \end{aligned}$$

If a nation has 2% annual growth rate, its population will double in 35 years.

Total Fertility Rates (TFR): It is one of the key measures of a nation's population growth. TFR is defined as the average number of children that would be born to a woman in her lifetime if the age-specific birth rates remain constant. The value of TFR varies from 1.9 in developed nations to 4.7 in developing nations. In 1950's the TFR has been 6.1. However, due to changes in cultural and technological set up of societies and government policies the TFR has come down which is a welcome change.

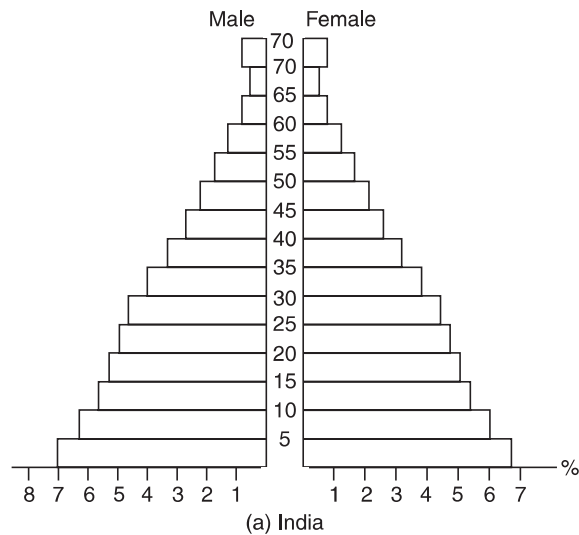
Infant mortality rate: It is an important parameter affecting future growth of a population. It is the percentage of infants died out of those born in a year. Although this rate has declined in the last 50 years, but the pattern differs widely in developed and developing countries.

Replacement level: This is an important concept in population dynamics or demography. Two parents bearing two children will be replaced by their offspring. But, due to infant mortality this replacement level is usually changed. For developing nations, where infant mortality is high and life expectancy is low, the replacement level is approx 2.7, whereas in developed nations it is 2.1.

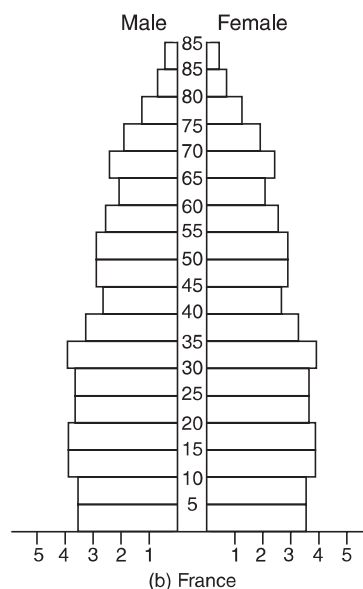
Age structure: Age structure of population of a nation can be represented by age pyramids, based upon people belonging to different age classes like pre-reproductive (0–14 years), reproductive (15–44 years) and post reproductive (45 years and above). We get three types of age pyramids:

(a) **Pyramid shaped:** Here the very young population is more, making a broad base and old people are less. This type indicates growing population. India, Bangladesh, Ethiopia, Nigeria are examples

of this type. The large number of individuals in very young age will soon enter into reproductive age, thus causing an increase in population, whereas less number of people in old age indicate less loss of population due to death (Fig. 7.2(a)).



(b) **Bell shaped:** It occurs in countries like France, USA and Canada where birth rates have in the past one or two decades declined resulting in people of almost equal number in age group 0–35 years. So in the next 10 years, the people entering into reproductive age group is not going to change much and such age-pyramids indicate stable populations (Fig. 7.2(b)).



(c) **Urn shaped:** Here number of individuals in very young class is smaller than the middle reproductive age class. In the next 10 years the number in reproductive age class will thus become less than before resulting in a decline of population growth. Germany, Italy, Hungary, Sweden and Japan are examples of this type (Fig. 7.2(c)).

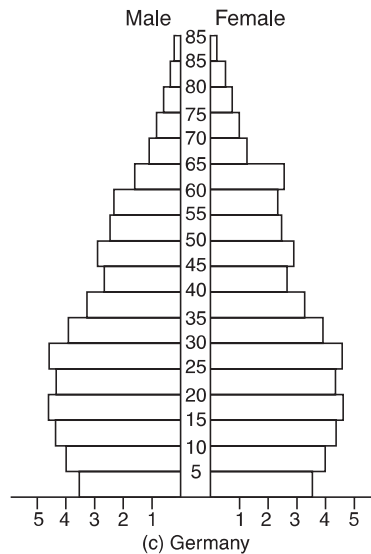


Fig. 7.2. Age pyramids (a) Pyramid-shaped expanding population—India, (b) Bell-shaped stable population—France, (c) Urn-shaped declining population—Germany.

(Source: UN Demography Year Book, 2000)

The TFR, age structure, infant mortality and replacement level are all important parameters determining population growth.

Case Studies

- Ethiopia is a developing nation with a pyramid shaped age structure indicating expanding population. Its TFR is 6.9 presently. Even if it aims to reach the replacement level by the year 2050, its population that is 57 million now would rise to 225 million by 2050 when TFR becomes 2.1 and continue growing until it levels off, 100 years later, at 370 million.
- Population growth is also affected due to AIDS in the HIV-prevalent countries mainly in Africa. The earlier population projections of UN are now found to be reduced by 8% in the seriously HIV-affected countries i.e. Mali, Rwanda, Uganda and Zambia. In Zimbabwe, HIV affects a quarter of the population aged between 15–49 yrs. In Botswana, 2/3rd of the 15 year olds are predicted to die of AIDS before reaching 50 years of age. About 30% of adult population in many African countries is HIV-positive. This has drastically reduced life expectancy in these countries.

Zero population growth (ZPG): When birth plus immigration in a population are just equal to deaths plus emigration, it is said to be zero population growth.

Male-female ratio: The ratio of boys and girls should be fairly balanced in a society to flourish. However, due to female infanticides and gender-based abortions, the ratio has been upset in many

countries including India. In China, the ratio of boys to girls became 140 : 100 in many regions which led to scarcity of brides.

Life expectancy: It is the average age that a newborn infant is expected to attain in a given country. The average life expectancy, over the globe, has risen from 40 to 65.5 years over the past century. In India, life expectancy of males and females was only 22.6 years and 23.3 years, respectively in 1900. In the last 100 years improved medical facilities and technological advancement has increased the life expectancy to 60.3 years and 60.5 years, respectively for the Indian males and females. In Japan and Sweden, life expectancy is quite higher, being 82.1–84.2 for females and 77–77.4 for males, respectively.

Demographic transition: Population growth is usually related to economic development. There occurs a typical fall in death rates and birth rates due to improved living conditions leading to low population growth, a phenomenon called demographic transition.

It is associated with urbanisation and growth and occurs in four phases:

(a) **Pre-industrial phase** characterized by high growth and death rates and net population growth is low.

(b) **Transitional phase** that occurs with the advent of industrialization providing better hygiene and medical facilities and adequate food, thereby reducing deaths. Birth rates, however, remain high and the population shows 2.5–3% growth rate.

(c) **Industrial phase** while there is a fall in birth rates thereby lowering growth rate.

(d) **Post industrial phase** during which zero population growth is achieved.

Demographic transition is already observed in most developing nations. As a result of demographic transition the developed nations are now growing at a rate of about 0.5% with a doubling time of 118 years. However, the matter of concern is that more than 90% of the global population is concentrated in developing nations which have a growth rate a little more than 2%, and a doubling time of less than 35 years.

(B) Population Explosion: There has been a dramatic reduction in the doubling time of the global human population, as we have already discussed. In the 20th century, human population has grown much faster than ever before. Between 1950–1990, in just 40 years the population crossed 5 billion mark with current addition of about 92 million every year, or so to say, adding a new Mexico every year. In the year 2000, the world population was 6.3 billion and it is predicted to grow four times in the next 100 years. This unprecedented growth of human population at an alarming rate is referred to as **population explosion**.

The Indian Scenario: India is the second most populous country of the world with 1 billion people. If the current growth rates continue, it will have 1.63 billion people by 2050 and will become the most populous country surpassing China. So we are heading for very serious ramifications of the population explosion problem. Do we have the resources and provisions for feeding, housing, educating and employing all those people being added every year? If we look at the population statistics of our country we find that in just 35 years after independence we added another India in terms of population. On 11th May, 2000 we became 1 billion and now we can say that every 6th person in this world is an Indian.

The Population Clock

Every second, on an average 4–5 children are born and 2 people die, thus resulting in net gain of nearly 2.5 person every second. This means that every hour we are growing by about 9000 and everyday by about 2,14,000.

Population explosion is causing severe resource depletion and environmental degradation. Our resources like land, water, fossil fuels, minerals etc. are limited and due to over exploitation these resources are getting exhausted. Even many of the renewable resources like forests, grasslands etc. are under tremendous pressure. Industrial and economic growth are raising our quality of life but adding toxic pollutants into the air, water and soil. As a result, the ecological life-support systems are getting jeopardized. There is a fierce debate on this issue as to whether we should immediately reduce fertility rates through worldwide birth control programs in order to stabilize or even shrink the population or whether human beings will devise new technologies for alternate resources, so that the problem of crossing the carrying capacity of the earth will never actually come.

There are two very important views on population growth which need a mention here:

Malthusian theory: According to Malthus, human populations tend to grow at an exponential or compound rate whereas food production increases very slowly or remains stable. Therefore, starvation, poverty, disease, crime and misery are invariably associated with population explosion. He believes “positive checks” like famines, disease outbreak and violence as well as “preventive checks” like birth control stabilize population growth.

Marxian theory: According to Karl Marx, population growth is a symptom rather than the cause of poverty, resource depletion, pollution and other social ills. He believed that social exploitation and oppression of the less privileged people leads to poverty, overcrowding, unemployment, environmental degradation that in turn, causes over population.

A compromise between the two views is required because all these factors seem to be interdependent and interrelated. Equity and social justice to all, allowing everyone to enjoy a good standard of living is the need of the hour that can voluntarily help in achieving a stabilized global population.

(C) Family Welfare Programmes: Population explosion is like a time bomb that must be diffused well in time. The population must be kept much below the carrying capacity and stabilized, so that the aftermath of explosion could be avoided.

It is not precisely known as to how long can we continue our exponential growth in population and resource use without suffering overshoot or dieback. We are getting warning signals that if not controlled, the increasing population is going to deplete all the resources beyond their regeneration capacity. A catastrophic doomsday model warns us that the earth cannot sustain more than two more doublings *i.e.* 25 billion.

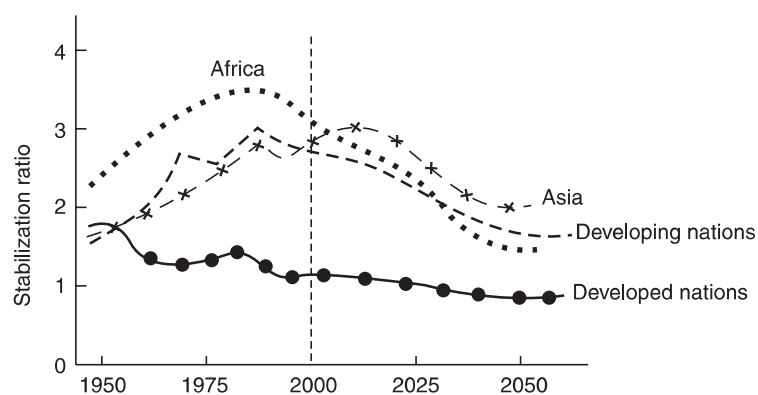


Fig. 7.3. Stabilization ratio of developing & developed nations, Africa and Asia. A ratio of 1 achieved in developed nations around 2000 indicates zero population growth in developed nations while Africa is presently having the highest ratio.

The United Nations projections about **population stabilization** of developed and developing nations and that of Asia are shown in Fig. 7.3. The population stabilization ratio is derived by dividing crude birth rate by crude death rate. As evident, developed nations have already achieved a stabilization ratio of 1 around the year 2000, which is more or less stabilized indicating zero population growth. Developing nations including Asia, on the other hand, is yet having a high stabilization ratio nearing 3, which is however, on a decline and is expected to lower down substantially by 2025. Stabilization in developing nations is possible only through various family welfare programmes.

The Kerala Model (A case study)

Kerala has earned the distinction of having lowest birth rates among all the states of India. The main parameters deciding the effectivity of this model depends upon the age of marriage for women at 21 years, as against Indian average of 18 years, female literacy of 53% against Indian average of 13%, greater emphasis on primary education with 60% budget provision for the same, as against 50% in many other states, better public distribution system of food among 97% of population, better medical facilities in rural areas and greater success of family planning programmes. The Kerala Model has its own success story emphasizing the effectivity of social justice approach for family planning.

(D) Family Planning: Family planning allows couples to decide their family size and also the time spacing of their offspring. Almost every culture in the past used to practise some traditional fertility control methods through some traditions, taboos and folk medicine.

Modern science has provided several birth control techniques including mechanical barriers, surgical methods, chemical pills and physical barriers to implantation. More than a hundred contraceptive methods are on trial. The **United Nations Family Planning Agency** provides funds to 135 countries. Many of these countries include abortion as a part of the population control programme which very often encourages female infanticide thereby disturbing the optimal male: female ratio in a society. The birth control programmes have often faced strong opposition from religious groups.

Nonetheless, **World Health Organization (WHO)** estimates that today about 50 percent of the worlds' married couples adopt some family planning measures as compared to just 10% about 30 years back. Still some 300 million couples do not have access to family planning.

The Indian Context

India started the family planning programme in 1952 while its population was nearly 400 million. In 1970's, forced family planning campaign by the Government resulted in a turmoil all over the country. In 1978, the government raised the legal minimum age of marriage from 18 to 21 for men and 15 to 18 years for women. Even in 1981 census no drop in population growth was observed. Since then funding for family planning programmes has been increased further.

Unable to reach a consensus regarding population policy, the state governments in 2000 were allowed to adopt their own approach. In Kerala, the population has been stabilized with a focus on social justice as already discussed. It is now comparable to many industrialized nations including USA and it has proved that wealth is not a pre-requisite for zero population growth. Andhra Pradesh has also just achieved the target of ZPG in 2001, but it has been done with a different approach. The poor class was encouraged to be sterilized after two children by paying cash incentives, better land, housing, wells and subsidized loans. In contrast, Bihar and U.P. have showed increase in their growth rates (more than 2.5%).

Successful family planning programme need significant societal changes including social, educational and economic status for women, social security, political stability, proper awareness and confidence building alongwith accessibility and effectivity of the birth control measures.

7.2 CLIMATE CHANGE

Climate is the average weather of an area. It is the general weather conditions, seasonal variations and extremes of weather in a region. Such conditions which average over a long period – at least 30 years – are called climate.

The **Intergovernmental Panel on Climate Change (IPCC)** in 1990 and 1992 published the best available evidence about past climate change, the green house effect and recent changes in global temperature. It is observed that earth's temperature has changed considerably during the geological times. It has experienced several glacial and interglacial periods. However, during the past 10,000 years of the current interglacial period the mean average temperature has fluctuated by 0.5–1°C over 100 to 200 year period. We have relatively stable climate for thousands of years due to which we have practised agriculture and increased in population. Even small changes in climatic conditions may disturb agriculture that would lead to migration of animals including humans.

Anthropogenic (man-made) activities are upsetting the delicate balance that has been established between various components of the environment. Greenhouse gases are increasing in the atmosphere resulting in increase in the average global temperature. This may upset the hydrological cycle, result in floods and droughts in different regions of the world, cause sea level rise, changes in agriculture productivity, famines and death of humans as well as livestock.

The global change in temperature will not be uniform everywhere and will fluctuate in different regions. The places at higher latitudes will be warmed up more during late autumn and winter than the places in tropics. Poles may experience 2 to 3 times more warming than the global average, while warming in the tropics may be only 50 to 100% on an average. The increased warming at poles will reduce the thermal gradient between the equator and high latitude regions decreasing the energy available to the heat engine that drives the global weather machine. This will disturb the global pattern of winds and ocean currents as well as the timing and distribution of rainfall. Shifting of ocean currents may change the climate of Iceland and Britain and may result in cooling at a time when rest of the world warms. By a temperature increase of 1.5 to 4.5°C the global hydrological cycle is expected to intensify by 5 to 10%. Disturbed rainfall will result in some areas becoming wetter and the others drier. Although rainfall may increase, higher temperatures will result in more evapotranspiration leading to annual water deficit in crop fields.

IPCC Report, 2007

The Report of the United Nations Intergovernmental Panel on Climate Change (IPCC), 2007 has been prepared by more than 3000 scientific expert researchers from various countries. The report highlights the unequal availability of water i.e. excess or lack of water leading to increase in droughts and floods. Glaciers in Himalayas will melt and the size and number of glacial lakes will increase. The mid latitude and semi arid regions of the world will experience drier years. Africa will experience water stress. There will be increased availability of water in moist tropics and high latitudes. Rain dependent agricultural produce will get a boost in North America. Sea level and human activities together will contribute to loss of coastal wetlands. Fresh water availability will decrease by 2050. More than a billion people will be at greater risk. The report assesses that 40 per cent species will become extinct. Human health will be affected. There will be increase in number of deaths, diseases like diarrhoea, cardiovascular diseases, etc.

The IPCC headed by Dr. R.K. Pachauri jointly with former US vice-president Al Gore was awarded Nobel Peace Prize for 2007.

7.3 GLOBAL WARMING

Troposphere, the lowermost layer of the atmosphere, traps heat by a natural process due to the presence of certain gases. This effect is called **Greenhouse Effect** as it is similar to the warming effect observed in the horticultural greenhouse made of glass. The amount of heat trapped in the atmosphere depends mostly on the concentrations of “heat trapping” or “greenhouse” gases and the length of time they stay in the atmosphere. The major greenhouse gases are carbon dioxide, ozone, methane, nitrous oxide, chlorofluorocarbons (CFCs) and water vapours. The average global temperature is 15°C. In the absence of greenhouse gases this temperature would have been –18°C. Therefore, Greenhouse Effect contributes a temperature rise to the tune of 33°C. Heat trapped by greenhouse gases in the atmosphere keeps the planet warm enough to allow us and other species to exist. The two predominant greenhouse gases are water vapours, which are controlled by hydrological cycle, and carbon dioxide, which is controlled mostly by the global carbon cycle. While the levels of water vapour in the troposphere have relatively

remained constant, the levels of carbon dioxide have increased. Other gases whose levels have increased due to human activities are methane, nitrous oxide and chlorofluorocarbons. Deforestation has further resulted in elevated levels of carbon dioxide due to non-removal of carbon dioxide by plants through photosynthesis.

Warming or cooling by more than 2°C over the past few decades may prove to be disastrous for various ecosystems on the earth including humans, as it would alter the conditions faster than some species could adapt or migrate. Some areas will become inhabitable because of drought or floods following a rise in average sea level.

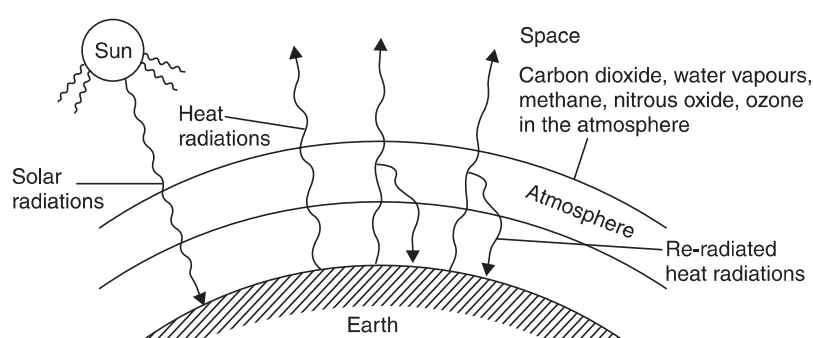


Fig. 7.4. The greenhouse effect.

Greenhouse gases

The phenomenon that worries the environmental scientists is that due to anthropogenic activities there is an increase in the concentration of the greenhouse gases in the air that absorb infrared light containing heat and results in the re-radiation of even more of the outgoing thermal infrared energy, thereby increasing the average surface temperature beyond 15°C. The phenomenon is referred to as the **enhanced greenhouse effect** to distinguish its effect from the one that has been operating naturally for millennia.

The greenhouse gases present in the troposphere and resulting in an increase in the temperature of air and the earth are discussed here:

Carbon dioxide

It contributes about 55% to global warming from greenhouse gases produced by human activity. Industrial countries account for about 76% of annual emissions. The main sources are fossil fuel burning (67%) and deforestation, other forms of land clearing and burning (33%). CO₂ stays in the atmosphere for about 500 years. CO₂ concentration in the atmosphere was 355 ppm in 1990 that is increasing at a rate of 1.5 ppm every year.

Chlorofluorocarbons (CFCs)

These are believed to be responsible for 24% of the human contribution to greenhouse gases. They also deplete ozone in the stratosphere. The main sources of CFCs include leaking air conditioners and refrigerators, evaporation of industrial solvents, production of plastic foams, aerosols, propellants etc. CFCs take 10–15 years to reach the stratosphere and generally trap 1500 to 7000 times more heat per

molecule than CO_2 while they are in the troposphere. This heating effect in the troposphere may be partially offset by the cooling caused when CFCs deplete ozone during their 65 to 110 years stay in the stratosphere. Atmospheric concentration of CFC is 0.00225 ppm that is increasing at a rate of 0.5% annually.

Methane (CH_4)

It accounts for 18% of the increased greenhouse gases. Methane is produced when bacteria break down dead organic matter in moist places that lack oxygen such as swamps, natural wetlands, paddy fields, landfills and digestive tracts of cattle, sheep and termites. Production and use of oil and natural gas and incomplete burning of organic material are also significant sources of methane. Methane stays in the atmosphere for 7–10 years. Each methane molecule traps about 25 times as much heat as a CO_2 molecule. Atmospheric concentration of methane is 1.675 ppm and it is increasing at a rate of 1% annually.

Nitrous Oxide (N_2O)

It is responsible for 6% of the human input of greenhouse gases. Besides trapping heat in the troposphere it also depletes ozone in the stratosphere. It is released from nylon products, from burning of biomass and nitrogen rich fuels (especially coal) and from the break down of nitrogen fertilizers in soil, livestock wastes and nitrate contaminated groundwater. Its life span in the troposphere is 140–190 years and it traps about 230 times as much heat per molecule as CO_2 . The atmospheric concentration of N_2O is 0.3 ppm and is increasing at a rate of 0.2% annually.

Impacts of Enhanced Greenhouse Effect

The enhanced greenhouse effect will not only cause global warming but will also affect various other climatic and natural processes.

(i) **Global temperature increase:** It is estimated that the earth's mean temperature will rise between 1.5 to 5.5°C by 2050 if input of greenhouse gases continues to rise at the present rate. Even at the lower value, earth would be warmer than it has been for 10,000 years.

(ii) **Rise in sea level:** With the increase in global temperature sea water will expand. Heating will melt the polar ice sheets and glaciers resulting in further rise in sea level. Current models indicate that an increase in the average atmospheric temperature of 3°C would raise the average global sea level by 0.2–1.5 meters over the next 50–100 years.

One meter rise in sea level will inundate low lying areas of cities like Shanghai, Cairo, Bangkok, Sydney, Hamburg and Venice as well as agricultural lowlands and deltas in Egypt, Bangladesh, India, China and will affect rice productivity. This will also disturb many commercially important spawning grounds, and would probably increase the frequency of storm damage to lagoons, estuaries and coral reefs.

In India, the Lakshadweep Islands with a maximum height of 4 meters above the sea level may be vulnerable. Some of the most beautiful cities like Mumbai may be saved by heavy investment on embankment to prevent inundation.

Life of millions of people will be affected by the sea level rise who have built homes in the deltas of the Ganges, the Nile, the Mekong, the Yangtze and the Mississippi rivers.

(iii) **Effects on human health:** The global warming will lead to changes in the rainfall pattern in many areas, thereby affecting the distribution of vector-borne diseases like malaria, filariasis, elephantiasis etc.

Areas which are presently free from diseases like malaria, schistosomiasis etc. may become the breeding grounds for the vectors of such diseases. The areas likely to be affected in this manner are Ethiopia, Kenya and Indonesia. Warmer temperature and more water stagnation would favour the breeding of mosquitoes, snails and some insects, which are the vectors of such diseases.

Higher temperature and humidity will increase/aggravate respiratory and skin diseases.

(iv) **Effects on agriculture:** There are different views regarding the effect of global warming on agriculture. It may show positive or negative effects on various types of crops in different regions of the world. Tropical and subtropical regions will be more affected since the average temperature in these regions is already on the higher side. Even a rise of 2°C may be quite harmful to crops. Soil moisture will decrease and evapotranspiration will increase, which may drastically affect wheat and maize production.

Increase in temperature and humidity will increase pest growth like the growth of vectors for various diseases. Pests will adapt to such changes better than the crops.

To cope up with the changing situation, drought resistant, heat resistant and pest resistant varieties of crops have to be developed.

Measures to Check Global Warming

To slow down enhanced global warming the following steps will be important:

- (i) Cut down the current rate of use of CFCs and fossil fuel.
- (ii) Use energy more efficiently.
- (iii) Shift to renewable energy resources.
- (iv) Increase nuclear power plants for electricity production.
- (v) Shift from coal to natural gas.
- (vi) Trap and use methane as a fuel.
- (vii) Reduce beef production.
- (viii) Adopt sustainable agriculture.
- (ix) Stabilize population growth.
- (x) Efficiently remove CO₂ from smoke stacks.
- (xi) Plant more trees.
- (xii) Remove atmospheric CO₂ by utilizing photosynthetic algae.

7.4 ACID RAIN

Oxides of sulfur and nitrogen originating from industrial operations and fossil fuel combustion are the major sources of acid forming gases. Acid forming gases are oxidised over several days by which time they travel several thousand kilometers. In the atmosphere these gases are ultimately converted into

sulfuric and nitric acids. Hydrogen chloride emission forms hydrochloric acid. These acids cause acidic rain. Acid rain is only one component of acidic deposition. Acidic deposition is the total of wet acidic deposition (acid rain) and dry deposition.

Rain water is turned acidic when its pH falls below 5.6 (Fig. 7.5). In fact clean or natural rain water has a pH of 5.6 at 20°C because of formation of carbonic acid due to dissolution of CO_2 in water.

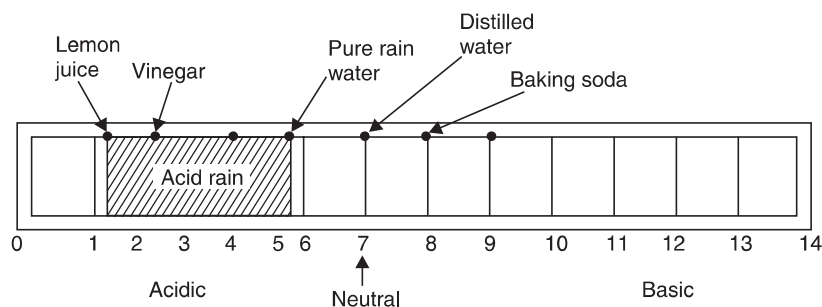


Fig. 7.5. The pH scale of common substances.

The Adirondack Lakes located in the state of New York are known to receive acid rains.

The strong acids like sulphuric acid (H_2SO_4) and nitric acid (HNO_3) dissolved or formed in rainwater dissociate or release hydrogen ions thereby increasing the acidity in rain drops.

Generally sulfuric acid forms a major fraction of acid rain, followed by nitric acid and a very small fraction of other acids. However, in urban areas calcium (Ca^{2+}), magnesium (Mg^{2+}) and ammonium (NH_4^+) ions help to neutralize the rain drops shifting the overall H^+ towards basic scale. The overall pH of any raindrop is due to the net effect of carbonic acid, sulfuric acid, nitric acid and other acidic constituents or any neutralizers such as ammonia.

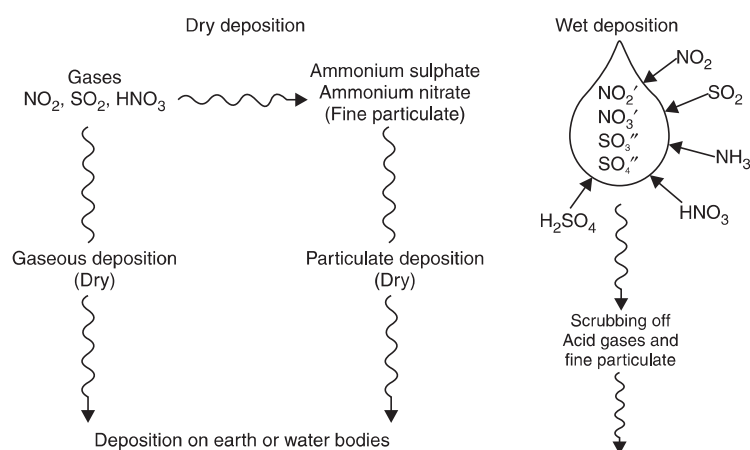


Fig. 7.6. Acid deposition (dry deposition and wet deposition).

In the absence of rain, dry deposition of acid may occur. Acid forming gases like oxides of sulphur and nitrogen and acid aerosols get deposited on the surface of water bodies, vegetation, soil and other materials. On moist surfaces or in liquids these acid forming gases can dissolve and form acids

similar to that formed in acid rain. If the oxidizers are present on the liquid surfaces then these gases undergo oxidation to form acids. Fine particles or acid droplets can act as nuclei for water to condense to form rain droplets. By such process sulfuric acid is incorporated into the droplets. In the clouds additional SO_2 and NO_2 contact the droplets and get absorbed which can be oxidized by the dissolved hydrogen peroxide (H_2O_2) or other oxidizers. In the droplets falling from the clouds additional acidic gases and aerosol particles get incorporated, further decreasing their pH. A unit decrease in pH value causes 10 times increase in acidity. Average pH in rainfall over eastern United States from April 1979 to March 1980 was less than 5.0. In India acid rain is recorded from certain places:

Name of place	pH of rainwater
Kodaikanal	5.18
Minicoy	5.52
Mohanbari	5.50

Effects of Acid Rain

Acid rain causes a number of harmful effects below pH 5.1. The effects are visible in the aquatic system even at pH less than 5.5.

- It causes deterioration of buildings especially made of marble *e.g.* monuments like Taj Mahal. Crystals of calcium and magnesium sulphate are formed as a result of corrosion caused by acid rain.
- It damages stone statues. Priceless stone statues in Greece and Italy have been partially dissolved by acid rain.
- It damages metals and car finishes.
- Aquatic life especially fish are badly affected by lake acidification.
- Aquatic animals suffer from toxicity of metals such as aluminium, mercury, manganese, zinc and lead which leak from the surrounding rocks due to acid rain.
- It results in reproductive failure and killing of fish.
- Many lakes of Sweden, Norway, Canada have become fishless due to acid rain.
- It damages foliage and weakens trees.
- It makes trees more susceptible to stresses like cold temperature, drought, etc. Many insects and fungi are more tolerant to acidic conditions and hence they can attack the susceptible trees and cause diseases.

Control of Acid Rain

- Emission of SO_2 and NO_2 from industries and power plants should be reduced by using pollution control equipments.
- Liming of lakes and soils should be done to correct the adverse effects of acid rain.
- A coating of protective layer of inert polymer should be given in the interior of water pipes for drinking water.

7.5 OZONE LAYER DEPLETION

For the last 450 million years the earth has had a natural sunscreen in the stratosphere called the ozone layer. This layer filters out harmful ultraviolet radiations from the sunlight and thus protects various life forms on the earth.

Ozone is a form of oxygen. The molecule of oxygen contains two atoms whereas that of ozone contains three (O_3). In the stratosphere ozone is continuously being created by the absorption of short wavelength ultraviolet (UV) radiations. Ultraviolet radiations less than 242 nanometers decompose molecular oxygen into atomic oxygen (O) by photolytic decomposition.



The atomic oxygen rapidly reacts with molecular oxygen to form ozone.



(M is a third body necessary to carry away the energy released in the reaction).

Ozone thus formed distributes itself in the stratosphere and absorbs harmful ultraviolet radiations (200 to 320 nm) and is continuously being converted back to molecular oxygen.



Absorption of UV radiations results in heating of the stratosphere.

The net result of the above reactions is an equilibrium concentration of ozone. Ozone concentration in about 24 km of the stratosphere *i.e.* from 16 km to 40 km away from earth is about 10 ppm (as compared to 0.05 ppm concentration of harmful tropospheric ozone). This equilibrium is disturbed by reactive atoms of chlorine, bromine etc. which destroy ozone molecules and result is thinning of ozone layer generally called ozone hole.

The amount of atmospheric ozone is measured by 'Dobson Spectrometer' and is expressed in **Dobson units (DU)**. One DU is equivalent to a 0.01 mm thickness of pure ozone at the density it would possess if it were brought to ground level (1atm) pressure. Normally over temperate latitude its concentration is about 350 DU, over tropics it is 250 DU whereas at subpolar regions (except when ozone thinning occurs) it is on an average 450 DU. It is because of the stratospheric winds which transport ozone from tropical towards polar regions.

Thinning of Ozone Layer

The Antarctic ozone hole was discovered by Dr. Joe C. Farman and his colleagues in the British Antarctic Survey who had been recording ozone levels over this region since 1957. During spring season of south pole *i.e.* September to November each year ozone depletion is observed. Steep decline has been observed since mid 1970s with a record low concentration of 90 DU in early October of 1993.

Chlorofluorocarbons (CFCs) are mainly responsible for ozone depletion in the stratosphere. CFCs are a group of synthetic chemicals first discovered by Thomas Midgley Jr. in 1930. CFC-11 and CFC-12 are the CFCs most commonly used. CFCs are used as coolants in refrigerators and air conditioners, as propellants, cleaning solvents, sterilant and in styrofoam etc. CFCs released in the troposphere reach the stratosphere and remain there for 65–110 years destroying O_3 molecules. In 1974, Rowland and Molina warned that CFCs are lowering the concentration of ozone in the stratosphere and

predicted severe consequences. It was however, in 1985 that scientists for the first time discovered that 50% (98% in some areas) of upper stratospheric ozone over Antarctica was destroyed during the Antarctic spring and early summer (September-December). At Antarctic region the temperature during winter drops to -90°C . The winds blowing in a circular pattern over earth's poles create polar vortices. Water droplets in clouds when they enter these vortices form ice crystals. CFCs get collected on the surfaces of these ice crystals and destroy ozone much faster. Similar destruction of ozone over North Pole occurs during Arctic spring and early summer (February-June). The depletion is 10–25% and it is less than that observed at south pole.

Nitrous oxide emitted by supersonic aircrafts during combustion of fossil fuel, and use of nitrogen fertilizers breaks ozone molecules. Chlorine liberated from chlorofluorocarbons also break ozone molecules. The chain reaction started in Antarctic spring *i.e.* August/September continues till nitrogen dioxide is liberated from nitric acid formed in the stratosphere by photolysis (breakdown by sunlight). Nitrogen dioxide combines with chlorine and stops further destruction of ozone.

Effects of Ozone Depletion

- Ozone depletion in the stratosphere will result in more UV radiation reaching the earth especially UV-B (290–320 nm). The UV-B radiations affect DNA and the photosynthetic chemicals. Any change in DNA can result in mutation and cancer. Cases of skin cancer (basal and squamous cell carcinoma) which do not cause death but cause disfigurement will increase.
- Easy absorption of UV rays by the lens and cornea of eye will result in increase in incidents of cataract.
- Melanin producing cells of the epidermis (important for human immune system) will be destroyed by UV-rays resulting in immuno-suppression. Fair people (who cannot produce enough melanin) will be at a greater risk of UV exposure.
- Phytoplanktons are sensitive to UV exposure. Ozone depletion will result in decrease in their population thereby affecting the population of zooplankton, fish, marine animals, in fact the whole aquatic food chain.
- Yield of vital crops like corn, rice, soybean, cotton, bean, pea, sorghum and wheat will decrease.
- Degradation of paints, plastics and other polymer material will result in economic loss due to effects of UV radiation resulting from ozone depletion.

7.6 URBANIZATION

An urban area is defined as a town or city along with its adjacent suburban fringes with a population exceeding 2500 (as per U.S. Bureau of Census). However, this value is set at a minimum of 10,000 – 50,000 in different countries. But in Denmark a population of 250 makes an area urban.

Urbanization is the growth of urban population by natural growth or immigration from rural areas. Rural population moves to urban areas to find employment there, because of poverty, declining agricultural jobs and inadequate and uneconomical market valuation of crops.

7.6.1 Characteristics and Patterns of Urban Population

- (i) These are heterogenous populations with much greater diversity of race, religion, ethnicity, socio-economic status and nationality as compared to rural population.
- (ii) In developing countries urban areas have more males, who migrate in search of job from rural areas. While in developed countries the women from rural areas migrate more to cities to seek employment.
- (iii) Urban population is growing very fast. Between 1850 and 2003, global urban population increased from a mere 2% to 47%, which will be 63% in 2050, as per UN (United Nations) projections.
- (iv) In 1900, there were only 19 cities having more than 1 million population and in 2003 the number of such cities increased to 400. These are called large cities.
Megacities, which have a population size exceeding 10 million have also increased in number from 8 in the year 1985 to 19 in the year 2003.
- (v) Urban growth is much faster in developing countries than in developed ones.
- (vi) Poverty is a main cause of urbanization, when poor people from villages move to urban areas and get crowded in slums with most unhygienic conditions.

7.6.2 Environment Related Urban Problems and Benefits

- (i) Urban areas suffer from serious problems of environmental pollution. There are high levels of air pollution and noise pollution due to industries and automobiles. Water is also polluted due to industrial wastewater discharge. Solid waste generation in urban areas is very high, and its proper disposal is a major problem.
- (ii) Proper sewerage facilities are often lacking in urban areas. In the crowded slums, the human waste is just deposited in gutters or vacant spaces, which become a breeding ground for pathogenic bacteria spreading salmonella and hepatitis infections.
In most of the cities heavy rainfall totally upsets the sewerage system.
- (iii) Cities are warmer than villages. Due to lot of heat released by various types of human activities in cities, which get retained by built structures and then slowly released into the atmosphere there is creation of **urban heat island**. This problem can be partially tackled by growing green belts of trees.
- (iv) Urbanization also provides several benefits when there is a well-planned city with proper transportation and residential facilities, reducing pollution problems.
- (v) Due to better access to health care, educational and social service facilities, infant mortality is less in cities.

7.7 AUTOMOBILE POLLUTION

Development of internal combustion engine (gasoline engine) in the late 1800s revolutionised modes of transportation. It was in 1900s that road transport progressed rapidly. In the last few decades the number of personal vehicles increased amazingly due to convenience, safety and affordability. This has come with a cost to the environment.

People specially in the urban areas are exposed to vehicular exhaust. The most vulnerable group is the traffic policemen. Pollutants from automobiles are emitted to the atmosphere through exhaust pipe (70%), crank case (20%) and by evaporation from fuel tank and carburettor (10%). The following pollutants are emitted from the vehicular exhaust. carbon monoxide (CO), unburnt hydrocarbons, oxides of nitrogen (NO_x), oxides of sulphur (SO_x), particulate matter, smoke and odour. The petrol engine exhaust contains carbon monoxide, unburnt hydrocarbons, nitrogen oxides and particulate matter. It is almost colourless. The diesel engine exhaust contains unburnt hydrocarbons, nitrogen oxides, sulphur oxides, particulate matter, smoke and odour. This exhaust is dirty and generally blue or black.

Emission of carbon monoxide and hydrocarbons is high during idling and deceleration and low during acceleration and high speed while levels of NO_x are high during acceleration and high speed and low during idling, deceleration and low speed. Levels of these pollutants are minimum at the recommended inform speed. Millions of vehicles spew into the atmosphere hundreds of metric tonnes of pollutants per day. Carbon monoxide contributes about 70% of the vehicular exhaust.

Effects of vehicular pollution: The ill effects of air pollutants have been discussed in the air pollution chapter. Vehicular pollution may produce from minor effects to serious illness sometimes leading to premature deaths. Vehicular pollution produces and aggravates respiratory and cardiovascular diseases. In the metros the situation is of special concern as thousands of people die prematurely due to this type of pollution. The worst affected cities are Delhi, Kolkata, Agra, Ahmedabad, Jaipur, Kanpur, Nagpur, Mumbai, Chennai etc. The pollution problems are aggravated in the cities where alongwith SO_2 , the particulate (suspended particulate matter) concentration is also high. This increases morbidity and mortality due to their synergistic (enhancing each others toxicity) effects.

Besides affecting human health, vehicular pollution affects environment also. Vehicular pollutants produce photochemical smog in the presence of sunlight. Los Angeles has a large number of vehicles and abundant sunshine and experiences frequent incidents of photochemical smog. For this reason it is called the pollution capital of the world. Vehicular pollution contributes to the major green house gas CO_2 which contributes to climate change (global warming). Vehicles are responsible for one-third of worlds total oil consumption resulting in depletion of this resource at a faster rate. Vehicular pollution seriously affects the living beings and crops especially when pollutants get trapped under inversion layer in winter months.

Control measures: Vehicular pollution can be minimized or controlled by adopting some of the following measures:

- Use of alternate fuel which generates less pollutants on combustion.
- Ensure complete combustion of fuel which changes pollutants into less harmful products.
- Use of catalytic converters.
- Modification of internal combustion engine for improvement of its efficiency.
- Replacement of internal Combustion engine with the engine which produces less pollutants.

7.8 ANIMAL HUSBANDRY

Animal husbandry or livestock raising is an important aspect of environment. It provides us food resources rich in protein. Sheep, pigs, chicken, turkeys, geese, ducks, cows, buffaloes and goats are some of the

important types of livestock. They provide us milk, eggs, meat and milk products. Besides this fish, crabs, prawns, shrimps, shellfish etc. also provide us protein rich food resources and are nowadays grown in aquaculture.

Some of the important environmental aspects associated with animal husbandry are discussed here:

(i) Domestication of animals leads to loss of *genetic diversity*. This is because animals with the most desirable traits are selected, which have value to humans. Other traits, which may be useful otherwise are not selected. This leads to uniform type of characters in the livestock population. If there is a disease outbreak in a uniform population, the whole population is susceptible and the loss is much greater as compared to a natural population, where some individuals would have genes to resist the pathogens due to genetic diversity.

(ii) The livestock have to be grazed on grassland or pastures. Very often the livestock grazing on a particular piece of grassland surpass its carrying capacity. *Carrying capacity* of any system is the maximum population that can be supported by it on a sustainable basis. When the grazing pressure is high, it leads to several problems like land degradation, soil erosion and loss of useful species.

Overgrazing removes the vegetation cover and exposes the soil to the action of strong wind and rainfall leading to soil erosion. Also, due to decline in litter, the organic matter and humus content of the soil decline leading to organically poor, dry, compacted soil. Overgrazing adversely affects the root stocks of good quality plants, which can no longer regenerate. Their position is gradually taken up by unpalatable, thorny plants, which are poor soil binders. Thus, there is overall degradation of land due to overgrazing.

(iii) For increasing livestock yields, use of hormones and antibiotics has come up, but the use of both is controversial, due to serious health concern for human consumers. Presence of implanted hormones in livestock products have been found to be very harmful to humans. Indiscriminate use of antibiotics in livestock results in evolution of such bacteria that have greater resistance to antibiotics, thus posing greater risks to us.

(iv) Animal wastes are rich in nitrogen and also phosphorus. If these wastes are not properly handled and managed these enter into water bodies with run-off, leading to over-nourishment or eutrophication of the same. The phenomenon has already been explained in an earlier chapter while dealing with effects of agriculture. Eutrophication leads to disastrous degradation of a water body in a short time.

(v) Animal wastes of certain livestock animals can serve as manure for increasing the productivity of croplands. They form an important component of organic farming.

(vi) Livestock are also a big contributor to emission of methane in the atmosphere. Food in the stomach of ruminants is acted upon by anaerobic bacteria and methane is produced. Methane comes to the atmosphere with their belch. Methane is 25 times more potent green house gas than CO₂ and contributes to climate change (global warming).

I. QUESTIONS

1. What do you mean by (a) Doubling time (b) total fertility rate (c) zero population growth (d) Life expectancy?
2. How can age-structure pyramids serve as useful tools for predicting population growth trends of a nation? Explain with examples.
3. What is meant by 'Population Explosion'? Discuss the Indian scenario.
4. What is meant by population stabilization? Discuss the family welfare and family planning program in Indian context.
5. Discuss the influence of environmental parameters and pollution on human health.
6. What are greenhouse gases and greenhouse effect ? Discuss the potential and contribution of these gases to global warming phenomenon.
7. What are the major implications of enhanced global warming?
What is meant by acid rain? How does it form? In which regions of India acid rain has been recorded?
8. What are the major impacts of acid rain and how can we control it?
9. Discuss the natural formation and occurrence of ozone in the stratosphere.
10. Which are the agents responsible for ozone depletion?
11. Discuss environment related problems and benefits.
12. Describe the effects of vehicular pollution on man and environment.
13. Discuss various environmental aspects associated with animal husbandry.

(A) FILL IN THE BLANKS

1. When a population increases by a fixed percentage it is called growth.
2. When birth plus immigration in a population are just equal to deaths plus emigration, it is known as
3. The phenomenon of fall in death rates and birth rates due to improved living conditions leading to low population growth is called
4. The average global temperature is 15°C. In the absence of greenhouse gases the temperature would have been
5. In acid rain, the pH of rain water falls below
6. The atmospheric emissions of NO₂ and cause acid rain.
7. Ozone layer acts as a natural sunscreen which protects life on this earth against rays.
8. Ozone depleting nature of CFC's was first reported by and
9. Cities are warmer than villages due to
10. CO contributes about % of vehicular exhaust.
11. Livestock contributes which is a greenhouse gas.

(B) CHOOSE THE CORRECT ANSWER

1. The world famous report on “The Limits to Growth” predicting that the world will meet the doom’s day, if growth continues limitlessly, was written by
(a) Myers *et al.* (b) Meadows *et al.*
(c) Brundtland (d) Wilson *et al.*
2. The present world population has just crossed
(a) 4 billion (b) 5 billion
(c) 6 billion (d) 8 billion.
3. If a nation has an annual growth rate of 2%, its population will double in
(a) 20 years (b) 35 years
(c) 50 years (d) 70 years.
4. Expanding population trend is predicted for the coming years when age-pyramid is
(a) Bell-shaped (b) Pyramid shaped
(c) Urn-shaped (d) None of these.
5. Every person in this world is an Indian.
(a) Fifth (b) Sixth
(c) Seventh (d) Tenth.
6. Which state in India has the lowest birth rate?
(a) Kerala (b) Bihar
(c) Jammu and Kashmir (d) Himachal Pradesh.
7. Which of the following gases has maximum contribution to enhanced greenhouse effect ?
(a) CFC’s (b) CH₄
(c) CO₂ (d) N₂O.
8. Cattle, sheep and termites are responsible for the release of the following greenhouse gas
(a) Methane (b) Carbon dioxide
(c) Nitrous oxide (d) All of these.
9. The most important agents for ozone depletion are
(a) Methane (b) CFC’s
(c) Nuclear fallout (d) Nitrous oxide.
10. Maximum depletion of ozone occurs on
(a) Equator (b) North pole
(c) South pole (d) Tropics.
11. Vehicular pollution has maximum effects on
(a) Respiratory system (b) Brain
(c) Kidney (d) Liver.
12. In the stomach of numinant which of the following bacteria produce methane?
(a) Aerobic bacteria (b) Anaerobic bacteria
(c) None of these (d) Both of these.

(C) WRITE TRUE OR FALSE

1. Need for more man-power in agriculture-based societies has been responsible for an exceptionally high population growth in developing countries. (True/False)
2. Total fertility rate and infant mortality rates are lower in developing countries. (True/False)
3. Life expectancy over the globe has improved over the past century. (True/False)
4. Green house effect is good while enhanced green house effect is bad. (True/False)
5. pH of normal rain is 5.6. (True/False)
6. Due to ozone depletion there will be increased incidence of skin cancer and eye cataract. (True/False)
7. Diesel engine exhaust contains sulphur oxides. (True/False)
8. Overgrazing can lead to soil erosion. (True/False)